

Question #1 of 77

Question ID: 1473654

At the inception of a market-rate plain vanilla swap, the value of the swap to the fixed-rate payer is:

- A) positive.
- B) zero.
- C) either positive or negative.

Chantal DuPont is the CFO of Vetements Verdun, a manufacturer of specialty clothing and uniforms, located in northern France. The firm is currently undergoing an expansion which will require DuPont to draw down 25 million on Vetements Verdun's credit line as a 90-day bridge loan before the mortgage closes. The money will not be needed for 60 days, at which point the interest rate will be determined. The interest rate on the loan will be based off 90-day LIBOR.

DuPont is becoming concerned because of signs that interest rates may begin to rise. The firm cannot afford to have its borrowing costs increase significantly over current rates. In response to DuPont's concerns, the company's CEO, Viviane Lamarre, has asked DuPont to hedge the firm's borrowing costs, even if that entails some near-term outlays.

DuPont and Lamarre discuss entering into a forward rate agreement (FRA) to hedge Vetements Verdun's interest rate exposure on the credit line. Current LIBOR rates are:

LIBOR rate	
30-day	2.6%
60-day	2.8%
90-day	3.0%
120-day	3.2%
150-day	3.3%
180-day	3.4%

They decide to go forward with the hedge and DuPont enters into the appropriate FRA for the full amount of 25 million.

In the first 30 days of the FRA, the fixed income markets rally sharply. The new set of LIBOR rates, on the thirtieth day of the FRA, is:

LIBOR rate	
30-day	2.2%
60-day	2.4%
90-day	3.6%
120-day	3.8%
150-day	3.8%
180-day	3.8%

At the settlement date, the interest savings on the loan term is 23,750. DuPont tells Lamarre, "I am looking forward to cashing our settlement check for 23,750." Lamarre adds, "Yes, and on top of that we get to borrow for 90 days at a below-market rate." Both DuPont and Lamarre are pleased with their decision to hedge.

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Question ID: 1586277

Which statement *most* accurately describes a 2 x 3 forward rate agreement?

- A) Contract expires in two months on an underlying loan settled in three months.
 - B) Two-month underlying interest rate on a contract settled in three months.
 - C) Underlying loan of two month maturity under a contract that expires in three months.
-

Question #3 - 5 of 77

Question ID: 1586278

Which forward rate agreement would *most* effectively hedge Vetements Verdun's exposure to LIBOR?

- A) 3 x 2.
 - B) 2 x 5.
 - C) 2 x 3.
-

Question #4 - 5 of 77

Question ID: 1586279

Which value is *closest* to the price of the most effective hedge for Vetements Verdun?

- A) 3.3%.
 - B) 3.6%.
 - C) 3.0%.
-

Question #5 - 5 of 77

Question ID: 1586280

What must the 90-day LIBOR rate have been at the expiration of the contract?

- A) 3.6%.
 - B) 4.0%.
 - C) 3.4%.
-

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Question ID: 1473669

The current U.S. dollar (\$) to Canadian dollar (C\$) exchange rate is 0.7. In a \$1 million currency swap, the party that is entering the swap to hedge existing exposure to C\$-denominated fixed-rate liability will:

- A) pay C\$1,428,571 at the beginning of the swap.
 - B) pay floating in C\$.
 - C) receive floating in C\$.
-

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Question ID: 1473641

Calculate the price (expressed as an annualized rate) of a 1x4 forward rate agreement (FRA) if the current 30-day rate is 5% and the 120-day rate is 7%.

- A) 7.47%.
- B) 6.86%.

C) 7.63%.

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Question ID: 1473672

Consider a one-year currency swap with semiannual payments. The payments are in U.S. dollars and euros. The current exchange rate of the euro is \$1.30 and interest rates are

	180 days	360 days
USD MRR	5.6%	6.0%
MRR	4.8%	5.4%

What is the fixed rate in euros?

- A) 5.318%.
 - B) 2.659%.
 - C) 5.245%.
-

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Question ID: 1473617

The contract price of a forward contract is:

- A) always the present value of the expected future spot price.
 - B) determined at the settlement date.
 - C) the price that makes the contract a zero-value investment at initiation.
-

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Question ID: 1473662

Writing a series of interest-rate puts and buying a series of interest-rate calls, all at the same exercise rate, is equivalent to:

- A) being the floating-rate payer in an interest rate swap.
- B) being the fixed-rate payer in an interest rate swap.

C) a short position in a series of forward rate agreements.

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Question ID: 1473625

The no-arbitrage price of a futures contract with a spot rate of 990, a time to maturity of 2 years, and a risk-free-rate of 5% is *closest* to:

- A) 1091
 - B) 1040
 - C) 792
-

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Question ID: 1473655

The fixed-rate on a semiannual 2-year interest rate swap is *closest* to the:

- A) coupon rate on a 2-year par bond with the same credit risk as the reference rate.
 - B) coupon rate on a 2-year par bond with the same credit risk as the fixed-rate payer.
 - C) current 180-day T-bill rate.
-

The Isle of Nefer is a developing country with its stock and futures markets enjoying record trading volumes due to the influx of foreign funds. You are looking to invest in the stock and futures markets in the Isle of Nefer. The representative stock market index, Nefer Industrial Index (NII), is currently priced at 8,765 and the one year NII future contract is currently trading at 8,920.

You have experience in using forward contracts but not futures. You discuss the possibility of investing in the Isle of Nefer using futures contract with your supervisor, Peter Filler, and he makes the following comments.

Comment 1: "A futures contract will have positive value after marking to market if the future price is up on that day."

Comment 2: "Given a quoted clean bond price of the CTD, when looking at a bond future the full price of the bond must be used which equals the clean price of the bond plus accrued interest. The futures price can then be calculated as:

$$QFP = \{(\text{full price}) \times (1 + R_f)^T + AI_T - FVC\}(1 / CF)$$

Where AI_T = accrued interest at futures maturity, R_f = risk-free rate, FVC = future value of coupon and CF = conversion factor

Peter Filler also suggests that you invest in Treasury bond futures. Exhibit 1 contains the relevant information.

Exhibit 1

Price of underlying deliverable 16 year 5% Treasury bond (just paid coupon) \$1,030

Expiration of Treasury bond futures contract 0.7 year

Conversion factor 1.08

Risk free rate 3.0 percent

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Question ID: 1473637

How will the price of the one-year stock index future perform over the next 12 months?

- A) The futures price will converge to the future spot index price, with the basis reducing to zero.
- B) The value will move approximately in line with the spot index price, with a fairly constant basis.
- C) The futures price will move approximately in line with the spot index price, though its actual level at the end of the year will depend more on supply and demand than on the spot price.

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Question ID: 1473638

Comment 1 is *best* described as:

- A) correct.

- B)** incorrect as the value of the futures contract should be negative after marking to market.
 - C)** incorrect as the value of the futures contract should be zero after marking to market.
-

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Question ID: 1473639

Comment 2 is *best* described as:

- A)** correct.
 - B)** incorrect as the full price should be the clean price less the current accrued interest.
 - C)** incorrect as the accrued interest at expiration should be deducted from the future value of full bond price in arriving at the quoted futures price.
-

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Question ID: 1473640

Using Exhibit 1, the quoted price of the Treasury bond futures contract should be *closest* to:

- A)** \$941.
 - B)** \$975.
 - C)** \$1,100.
-

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Question ID: 1473673

90 days ago the exchange rate was USD 0.83 per CDN and the term structure was:

	180 days	360 days
USD MRR	5.2%	5.6%
CDN MRR	4.8%	5.4%.

A 1 year, semi-annual settlement, fixed for fixed swap was initiated with 5.30% fixed for CDN and 5.52% fixed for USD on a principal of USD 1 million.

Current exchange rate is USD 0.84 per CDN and the yield curve is:

	90 days	270 days
USD MRR	5.2%	5.6%
Disc Factor	0.98717	0.95969
CDN MRR	4.8%	5.4%
Disc factor	0.98814	0.96108

What is the value of the swap to the USD interest payer?

- A) \$11,500.
 - B) \$10,126.
 - C) -\$3,472.
-

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Question ID: 1473627

To initiate an arbitrage trade if the futures contract is underpriced, the trader should:

- A) borrow at the risk-free rate, short the asset, and sell the futures.
 - B) short the asset, invest at the risk-free rate, and buy the futures.
 - C) borrow at the risk-free rate, buy the asset, and sell the futures.
-

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Question ID: 1473665

If the one year spot rate is 5%, the two-year spot rate is 5.5%, and the three year spot rate is 6%, the fixed rate on a 3-year annual pay swap is *closest* to:

- A)** 1.99%.
 - B)** 4.50%.
 - C)** 5.65%.
-

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Question ID: 1586281

30 days ago, J. Klein took a short position in a \$10 million (3X6) forward rate agreement (FRA) based on MRR and priced at 5%. The current MRR curve is:

- 30-day = 4.8%
- 60-day = 5.0%
- 90-day = 5.1%
- 120-day = 5.2%
- 150-day = 5.4%

The current value of the FRA, to the short, is *closest* to:

- A)** -\$15,154.
 - B)** -\$15,495.
 - C)** -\$15,280.
-

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Question ID: 1473661

The fixed-rate payer in an interest-rate swap has a position equivalent to a series of:

- A)** long interest-puts and short interest-rate calls.
 - B)** long interest-rate puts and calls.
 - C)** short interest-rate puts and long interest-rate calls.
-

Craig Champion, CFA, manages portfolios of U.S. securities for European investors. His clients have each hold different kinds of securities, and each has differing views with respect to hedging exchange rate risk.

Francois Levisque is a Belgian investor who holds a large diversified portfolio of U.S. equities. Levisque has a reputation for some success in timing the U.S. equity market. For example, he has often locked in gains on his portfolio with derivatives shortly before a market correction. Sometimes he also hedges his portfolio's currency risk.

Levisque has just instructed Champion to take a large short position in S&P 500 index, either with futures or with a forward contract. Champion notices that the futures price is less than the current spot price and consults with his colleague Danielle Silvers, CFA. Champion says he thinks that the futures price is less than the spot price because the dividend yield of the S&P 500 is greater than the Treasury Bill rate. Silvers says that it could just be backwardation.

Silvers also notes that the use of a forward contract might be a good idea because the contract will not attract the attention of other market participants who might react to Levisque's move. Champion tells Silvers that the reason Levisque wants to hedge his equity position is that he thinks all U.S. interest rates will increase soon. This, he believes, is bearish for equities.

Ragnar Hvammen is a Norwegian investor with a large investment in oil-related assets that he often hedges with futures contracts. Champion notices that the price of an oil futures contract is usually higher than the spot price. Hvammen uses short-term borrowings in dollars, from both European and U.S. banks, to meet the liquidity needs of his oil investments, and he has Champion hedge these loan positions with Eurodollar futures.

Silvers suggests that Champion should consider using T-bill futures to hedge the loans from U.S. banks, and use Eurodollar futures only for the Eurodollar loans. Champion says he will look into that, as well as forward rate agreements, as alternative hedging tools for Hvammen.

Champion is also evaluating pricing of Euro-bund futures. Specifically, he is looking for pricing on a 1.2-year contract. The CTD is a 2.5% 10-year, semi-annual coupon bond issued 1 year ago (just paid coupon) currently quoted at €104.10. The conversion factor for the bond is 1.08. At contract expiration, the underlying will have accrued interest of €0.42. Assume that the risk-free rate over the contract period is 1%.

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Question ID: 1586271

Champion and Silvers each gave a reason for why the futures price of the S&P 500 index might be less than the spot price. With respect to their statements, it is *most accurate* to conclude that:

A) neither statement is valid.

- B) Champion's statement is invalid while Silver's statement is valid.
 - C) both statements are valid.
-

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Question ID: 1586272

For a futures contract on an asset with no storage costs, convenience yield, or other expected cash flows over the term of the contract, there should be a:

- A) positive correlation between the futures price and interest rates and a negative correlation between the futures price and the spot price.
 - B) negative correlation between the futures price and interest rates and a positive correlation between the futures price and the spot price.
 - C) positive correlation between the futures price and both interest rates and the spot price.
-

Question #24 - 25 of 77

Question ID: 1586273

Oil futures prices might be higher than the spot price because:

- A) of reverse contango.
 - B) there are more costs than benefits to holding the asset.
 - C) there are more benefits than costs to holding the asset.
-

Question #25 - 25 of 77

Question ID: 1586274

The no-arbitrage futures price of the Euro-bond contract is *closest* to:

- A) €94.83.
 - B) €102.85.
 - C) €110.61.
-

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Question ID: 1473656

A plain vanilla interest-rate swap to the fixed-rate payer is equivalent to issuing a fixed-rate bond and:

- A)** buying a floating-rate bond.
 - B)** selling a series of interest rate calls.
 - C)** selling a series of interest rate puts.
-

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Question ID: 1508677

Consider a fixed-for-fixed 1-year \$100,000 semiannual currency swap with rates of 5.0% in USD and 4.8% in CHF, originated when the exchange rate is \$0.34. After the first settlement, the exchange rate is \$0.35 and the term structure is:

	90 days	180 days
MRR	5.2%	5.6%
Swiss	4.8%	5.4%

What is the value of the swap to the USD payer?

- A)** \$2,814.
 - B)** \$2,937.
 - C)** -\$2,719.
-

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Question ID: 1473658

For a 1-year quarterly-pay swap, an equivalent position with short puts and long calls would involve:

- A)** three put-call combinations expiring on the first three settlement dates of the swap.
 - B)** put-call combinations expiring on each of the four settlement dates.
 - C)** three put-call combinations on the last three settlement dates of the swap.
-

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Question ID: 1473652

The price and value of a plain vanilla interest-rate swap are:

- A)** never equal.
 - B)** only equal at the inception of a swap contract.
 - C)** equal in equilibrium.
-

Question #30 of 77

Question ID: 1473681

Consider a fixed-rate semiannual-pay equity swap where the equity payments are the total return on a \$1 million portfolio and the following information:

- 180-day MRR is 5.2%
- 360-day MRR is 5.5%
- Dividend yield on the portfolio = 1.2%

What is the fixed rate on the swap?

- A)** 5.4197%.
 - B)** 5.1387%.
 - C)** 5.4234%.
-

Question #31 of 77

Question ID: 1473616

At expiration, the value of a forward contract is:

- A)** always greater than or equal to zero.
 - B)** the difference between the contract price and the market value of the underlying asset.
 - C)** equal to the market price of the underlying asset.
-

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Question ID: 1473624

The value of a futures contract between the times when the account is marked-to-market is:

- A)** never less than the value of a forward contract entered into on the same date.
- B)** equal to the difference between the price of a newly issued contract and the settle price at the most recent mark-to-market period.
- C)** the same as the contract price.

Elodie Brodeur works in the finance department of a large fashion house in France. The international catwalk season will start in two months' time and Brodeur has worked out that the company will need a 3-month loan of €4m in two months' time. The company's lenders are typically retail banks offering loans at MRR plus 40 basis points. Brodeur is concerned that interest rates may rise during the next two months and wants to use a FRA to lock-in the borrowing cost. Brodeur has collected the rate information shown in Exhibit 1.

Exhibit 1: Current MRR Rates

60 day MRR 2.0%

90 day MRR 2.4%

150 day MRR 2.6%

210 day MRR 2.9%

One month after the initiation of the FRA the MRR rates are shown in Exhibit 2.

Exhibit 2: MRR Rates One Month Later

30 day MRR 1.8%

90 day MRR 2.5%

120 day MRR 2.8%

180 day MRR 3.0%

At the expiration of the FRA, 90-day MRR is 3.4%.

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Question ID: 1473645

Which of the following comments relating to Brodeur's use of a forward rate agreement is *least* accurate?

- A)** A short FRA can be used to lock into a fixed rate of borrowing commencing in two months' time and expiring in five months' time.
 - B)** The use of a FRA to hedge interest rate risk would lock Brodeur into paying a fixed rate plus 40 basis points for her borrowing.
 - C)** The use of a FRA to hedge interest rate risk on her future loan will mean that she no longer benefits if interest rates fall.
-

Question #34 - 36 of 77

Question ID: 1473646

Using the data in Exhibit 1, which of the following is *closest* to the forward price of the FRA?

- A)** 2.4%.
 - B)** 2.8%.
 - C)** 3.0%.
-

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Question ID: 1473647

For this question only, assume that the forward price of the FRA was 2.9%. Which of the following is the closest to the value accrued on the FRA from Brodeur's perspective one month after initiation of the contract?

- A)** +€2,300.
 - B)** +€2,200.
 - C)** -€2,265.
-

Question #36 - 36 of 77

Question ID: 1473648

Assuming the forward price of the FRA was 2.9%, which of the following is *closest* to the value of the FRA at expiration?

- A) €4,958.
 - B) €5,000.
 - C) €19,831.
-

Question #37 of 77

Question ID: 1473674

In an equity return swap for MRR, if the return on the underlying equity portfolio is negative for a payment period, the equity return payer will:

- A) receive a net payment greater than the loss of value on the equity portfolio.
 - B) make a net payment greater than the loss of value on the equity portfolio.
 - C) receive a net payment less than the loss of value on the equity portfolio.
-

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Question ID: 1473632

A stock is currently priced at \$110 and will pay a \$2 dividend in 85 days and is expected to pay a \$2.20 dividend in 176 days. The no arbitrage price of a six-month (182-day) forward contract when the effective annual interest rate is 8% is *closest* to:

- A) \$110.00.
 - B) \$110.06.
 - C) \$110.20.
-

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Question ID: 1473635

Consider a 9-month forward contract on a 10-year 7% Treasury note just issued at par. The effective annual risk-free rate is 5% over the near term and the first coupon is to be paid in 182 days. The price of the forward is *closest* to:

- A) 965.84.

B) 1,037.27.

C) 1,001.84.

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Question ID: 1473683

Consider a 1-year, \$5 million semiannual-pay fixed-rate equity swap initiated when the equity index is 750 and swap fixed rate is 3.7%. Equity index was at 760 at first settlement. It is now 270 days since inception of the swap and the index is at 767, 90-day MRR is 3.4% ($DF = 0.99157$) and 270-day MRR is 3.7% ($DF = 0.9730$). What is the value of the swap to the fixed-rate payer?

A) \$3,478.

B) -\$2,726.

C) -\$3,520.

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Question ID: 1473629

The value of the S&P 500 Index is 1,260. The continuously compounded risk-free rate is 5.4% and the continuous dividend yield is 3.5%. Calculate the no-arbitrage price of a 160-day forward contract on the index.

A) \$562.91.

B) \$1,270.54.

C) \$1,310.13.

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Question ID: 1586275

What is the value of a 6.00% 1x4 (30 days x 120 days) forward rate agreement (FRA) with a principal amount of \$2,000,000, 10 days after initiation if the MRR curve at that time reflects the following:

110-day MRR = 6.15% and

20-day MRR = 6.05%?

- A)** \$745.76.
 - B)** \$767.40.
 - C)** \$700.00.
-

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Question ID: 1473620

The theoretical price of a forward contract:

- A)** equals the long's expectation of the future price of the underlying asset.
 - B)** is the no-arbitrage price.
 - C)** is always greater than the current price of the underlying asset.
-

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Question ID: 1473668

A U.S. firm (U.S.) and a foreign firm (F) engage in a 3-year, annual pay currency swap; The USD fixed rate at initiation was 5% while FC fixed rate was 4%. At the beginning of the swap, \$2 million was paid by the U.S. firm and the exchange rate was 2 FC units per \$1. At the end of the swap period the exchange rate was 1.75 FC units per \$1.

At the end of year 1, firm:

- A)** U.S. pays firm F 160,000 FC units.
 - B)** F pays firm U.S. \$200,000.
 - C)** U.S. pays firm F \$200,000.
-

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Question ID: 1473663

Which of the following is *equivalent* to a pay-fixed swap with a tenor of two years with semi-annual swap payments and a fixed rate of 6% (exchanged for MRR)? The notional principal is \$100,000,000.

- A)** A strip of three forward rate agreements, which obligates the party to pay a fixed rate of 6% and receive six-month MRR on a notional principal of \$100,000,000.
 - B)** A forward rate agreement, which obligates the party to pay a fixed rate of 6% and receive six-month MRR on a notional principal of \$100,000,000.
 - C)** A strip of two forward rate agreements, which obligates the party to pay a fixed rate of 6% and receive six-month MRR on a notional principal of \$100,000,000.
-

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Question ID: 1473634

Calculate the price of a 200-day forward contract on an 8%, semi-annual, U.S. Treasury bond with a spot price of \$1,310. Next coupon payment will be made in 150 days. The annual risk-free rate is 5%.

- A)** \$1,333.50.
 - B)** \$1,270.79.
 - C)** \$1,305.22.
-

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Question ID: 1473615

The price of a forward contract:

- A)** depends on forward interest rates.
 - B)** is determined at contract initiation.
 - C)** changes over the term of the contract.
-

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Question ID: 1473667

Which of the following is equivalent to a pay-floating USD receive-fixed EUR currency swap position?

- A) A short position in a EUR bond coupled with the issuance of a USD-denominated floating rate note.
 - B) A long position in a EUR bond coupled with the issuance of a USD-denominated floating rate note.
 - C) A short position in a EUR bond coupled with a long position in a USD-denominated floating rate note.
-

Question #49 of 77

Question ID: 1473660

Which of the following is *equivalent* to a plain vanilla receive-fixed interest rate swap?

- A) A short position in a bond coupled with the issuance of a floating rate note.
 - B) A short position in a bond coupled with a long position in a floating rate note.
 - C) A long position in a bond coupled with the issuance of a floating rate note.
-

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Question ID: 1473628

Over the life of a swap, the price of the swap:

- A) is approximately equal to the market value of the swap.
 - B) fluctuates with changes in the yield curve.
 - C) does not change.
-

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Question ID: 1587862

The fixed-rate receiver in a plain vanilla interest rate swap has a position equivalent to a series of:

- A) short interest-puts and long interest-rate calls.
 - B) long interest-rate puts and short interest-rate calls.
 - C) long interest-rate puts.
-

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Question ID: 1473684

Consider a 1-year semiannual equity swap based on an index at 985 and a fixed rate of 4.4%. 90 days after the initiation of the swap, the index is at 982 and MRR is 4.6% for 90 days and 4.8% for 270 days. The value of the swap to the equity payer, based on a \$2 million notional value is *closest* to:

- A) -\$22,564
 - B) \$22,314
 - C) \$22,564
-

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Question ID: 1473621

At contract initiation, the value of a forward contract:

- A) is typically zero regardless of the price of the underlying asset.
 - B) is set to 100 by convention.
 - C) depends on the market price of the underlying asset.
-

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Question ID: 1473619

The price of a forward contract:

- A) is equal to the value of the contract in equilibrium.
 - B) is the settlement price for the underlying asset.
 - C) must be equal to the market price at contract termination.
-

Abel Smith works in the Treasury Department of OTS Ltd. OTS is an international construction firm, based in the United States. OTS hopes to raise €100 million through the issuance of a €100 million one-year fixed rate bond but is concerned about the currency risk exposure.

TNA Bank proposed a one year EUR-USD currency swap with semi-annual settlements to OTS to mitigate the exchange rate risk. The notional principal would be €100 million. The

bank provides the following information:

Exhibit 1: MRR Spot Rate (annualized)

Days	\$ MRR	PV Factors	Days	MRR	PV Factors
180	0.70%	0.9965	180	1.50%	0.9926
360	1.00%	0.9901	360	2.00%	0.9804

Using the information in Exhibit 1, the bank calculates that the currency swap's fixed rates are 0.85% on the USD and 1.75% on the euro.

The term structure of MRR and MRR spot rates three months after the swap initiation is shown in Exhibit 2:

Exhibit 2: MRR Spot Rate (annualized)

Days	\$ MRR	PV Factors	Days	MRR	PV Factors
90	0.85%	0.9979	90	1.30%	0.9968
270	1.20%	0.9911	270	2.30%	0.9830

Exhibit 3: Exchange Rates

Time	Exchange Rate	Time	Exchange Rate
Swap initiation	€1: USD1.43	1st settlement date	€1: USD1.55
90 days after swap initiation	€1: USD1.48	2nd settlement date	€1: USD1.55

OTS has an investment portfolio with similar weighting as the S&P500. Smith believes that the U.S. equity market could suffer further declines and OTS could hedge the equity risk using an equity swap. Smith obtained the outlook of the U.S. equity market in Exhibit 4.

Exhibit 4: Performance of S&P500 for the Next Three Quarters

Time	S&P Index Level
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Swap initiation	1,190
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1st quarter	1,000
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2nd quarter	980
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3rd quarter	1,050
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4th quarter	1,130
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Smith is proposing for OTS to be the party paying the equity return on a USD500 million one-year equity swap. OTS will be receiving fixed rate of 1% on a semi-annual basis.

Smith makes two comments:

Comment 1: A fixed-rate payer in an equity swap would not have to pay more than the fixed rate each period.

Comment 2: Compared to an interest rate swap, the first payment in an equity swap will not be known at initiation.

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Question ID: 1473677

Has the bank correctly calculated the fixed rates on the currency swap?

- A)** Both fixed rates are incorrectly calculated.
 - B)** One of the two fixed rates is incorrectly calculated.
 - C)** Both fixed rates are correctly calculated.
-

Question #56 - 58 of 77

Question ID: 1496747

If OTS decides on a fixed for fixed currency swap, what is the market value of the swap to OTS three months after swap initiation?

- A) The value of the euro payments is €97.68m and converting the euro value to USD using €1: USD1.55, the swap has a positive to OTS.
- B) The value of the euro payments is €100.27m and converting the euro value to USD using €1: USD1.48, the swap has a positive to OTS.
- C) The value of the euro payments is €101.8m and converting the euro value to USD using €1: USD1.43, the swap has a positive to OTS.
-

Question #57 - 58 of 77

Question ID: 1473679

Using the information in Exhibit 2, what is the market value of the equity swap to OTS three months after swap initiation?

- A) +USD80.35 million.
- B) +USD85.33 million.
- C) +USD88.76 million.
-

Question #58 - 58 of 77

Question ID: 1473680

How many of Smith's comments are correct?

- A) Neither comment is correct.
- B) One comment is correct.
- C) Both comments are correct.
-

Question #59 of 77

Question ID: 1473666

A \$10 million 2-year semi-annual-pay MRR-based interest-rate swap was initiated 180 days ago when swap fixed rate was 3.8%. The fixed rate on the swap is now 3.4% and the term structure is as follows:

Days	MRR	Discount Factor
180	3.00%	0.98522
360	3.20%	0.96899
540	3.40%	0.95148
720	4.00%	0.92593

Value of the swap to the payer is *closest* to:

- A) -\$45,633.
 - B) -\$58,114.
 - C) -\$29,229.
-

Question #60 of 77

Question ID: 1473653

The price of an interest rate swap is the:

- A) cost to purchase a swap.
 - B) fixed rate of interest.
 - C) market value of the swap.
-

Question #61 of 77

Question ID: 1473670

In which type of swap contract is notional principal *most likely* to be exchanged at initiation?

- A) Currency swap.
 - B) Equity swap.
 - C) Interest rate swap.
-

Question #62 of 77

Question ID: 1473675

An equity portfolio manager who has a positive long-term outlook for equities, but expects equity prices to decline over the next three months, would *most appropriately* enter into:

- A) a short straddle with an equity index as the underlying.
 - B) an equity swap as the equity return receiver.
 - C) an equity swap as the equity return payer.
-

Question #63 of 77

Question ID: 1473623

What is the difference between spot and futures prices? Spot prices are always:

- A) delivered to meet the futures obligation at expiration.
 - B) equal to the futures price at futures expiration.
 - C) lower than futures prices.
-

Question #64 of 77

Question ID: 1473626

During the life of a forward contract, the value of the contract is *best* described as:

- A) the difference between the spot price and the present value of the forward price of the underlying asset.
 - B) the difference between the future value of the spot price and the expected future price of the underlying asset.
 - C) the present value of the expected future price of the underlying asset.
-

Question #65 of 77

Question ID: 1586282

The floating-rate payer in a simple interest-rate swap has a position that is equivalent to:

- A) a series of long forward rate agreements (FRAs).
- B) a series of short FRAs.

C) issuing a floating-rate bond and a series of long FRAs.

Question #66 of 77

Question ID: 1473682

Consider a fixed-rate semiannual-pay equity swap where the equity payments are the total return on a \$1 million portfolio and the following information:

- 180-day MRR is 4.2%
- 360-day MRR is 4.5%
- Div. yield on the portfolio = 1.2%

What is the fixed rate on the swap?

- A) 4.3232%.
- B) 4.5143%.
- C) 4.4477%.
-

Question #67 of 77

Question ID: 1473643

A company has chosen to use a 6 x 9 FRA expiring in 6 months to mitigate the risk of paying a floating coupon on the bond issue. The current term structure for MRR is as follows:

Term	Interest Rate
180 days	5.65%
270 days	5.95%

What is the price of this forward rate agreement (FRA)?

- A) 3.19%
- B) 6.37%
- C) \$6.37
-

Question #68 of 77

Question ID: 1473631

Calculate the no-arbitrage forward price for a 90-day forward on a stock that is currently priced at \$50.00 and is expected to pay a dividend of \$0.50 in 30 days and a \$0.60 in 75 days. The annual risk free rate is 5% and the yield curve is flat.

- A) \$48.51.
 - B) \$50.31.
 - C) \$49.49.
-

Question #69 of 77

Question ID: 1473642

The price of a 3×5 forward rate agreement (FRA) is the:

- A) 2-month implied forward rate 3 months from today.
 - B) 3-month implied forward rate 5 months from today.
 - C) 2-month implied forward rate 5 months from today.
-

Question #70 of 77

Question ID: 1473657

A swap is equivalent to a series of:

- A) FRAs priced at market rates.
 - B) off-market FRAs.
 - C) interest rate calls.
-

John Williams, CFA, works in the treasury department of Sam Smith Leisure Inc., a U.S. based manufacturer of gym equipment. Recently he has been considering using derivative instruments to lock in returns on excess cash flows that tend to accumulate in the final quarter of each year as demand for equipment peaks during that time.

He estimates that this year, in 60-days, the company will have \$28.5 million in excess funds to invest for 90 days.

Williams is presenting to the board 60 days before the excess funds need to be deposited, which is also 30 days before the year end. He intends to suggest an FRA as a method of

locking in a return on the deposit. He intends to make the following two statements in favor of using an FRA.

Statement 1

As we are depositing cash, committing to an FRA will generate a cash inflow on the date we enter into it.

Statement 2

If rates move in our favor, we will receive a cash payment at the end of the notional borrowing period.

Williams will present the hypothetical rates and MRR shown in Exhibit 1 to illustrate the result of using an FRA. Current 2x5 FRA price is 3.8%. All rates are annualized.

Exhibit 1 – FRA Price and Theoretical Future MRR rates

Predicted MRR rates	In 30 days	In 60 days	In 90 days	In 120 days	In 150 days
30-day MRR	3.9%	4.0%	4.2%	4.4%	4.5%
60-day MRR	4.1%	4.4%	4.5%	4.7%	4.8%
90-day MRR	4.2%	4.7%	4.8%	4.9%	5.2%
120-day MRR	4.5%	5.0%	5.2%	5.3%	5.5%
150-day MRR	4.8%	5.3%	5.4%	5.6%	5.9%

One key question that the CFO is likely to ask is the predicted value of the FRA at the year end.

Finally Williams is to investigate the potential for Sam Smith Inc. to use a currency swap to borrow and invest in a manufacturing facility in Europe. A bank has offered the company a fixed for fixed currency swap involving US dollars and Euros.

Some of the swap details are outlined in Exhibit 2.

Exhibit 2– Currency Swap

Initiation:	1st January
Spot rate at initiation:	USD/EUR 1.19
Settlement:	Quarterly

Principal:	USD 40,000,000
Fixed EUR rate:	1.5%
Fixed USD rate:	1.3%

Question #71 - 74 of 77

Question ID: 1586284

Which of Williams' statements regarding FRAs is *most likely* correct?

- A) Only Statement 2 is correct.
 - B) Neither statement is correct.
 - C) Only Statement 1 is correct.
-

Question #72 - 74 of 77

Question ID: 1586285

Using the price and predicted MRR rates in Exhibit 1, which of the following is closest to the predicted value of the FRA at the year end?

- A) -\$100,000
 - B) --\$85,000
 - C) -\$62,000
-

Question #73 - 74 of 77

Question ID: 1586286

Using the details shown in Exhibit 2, under the terms of the currency swap at the first settlement date Sam Smith would *most likely*:

- A) pay EUR 150,000.
 - B) pay USD 130,000.
 - C) pay EUR 126,050.
-

Question #74 - 74 of 77

Question ID: 1586287

Which of the following statements regarding cash flows at the final settlement date for the currency swap outlined in Exhibit 2 is *most likely* correct?

- A) Without knowing the spot rates on the final settlement date, it is impossible to state the cash flows that occur.
 - B) Sam Smith Inc. will receive USD 40,000,000 plus the USD interest payment.
 - C) Sam Smith will pay USD 40,000,000 and receive the final USD interest payment.
-

Question #75 of 77

Question ID: 1473630

An index is currently 965 and the continuously compounded dividend yield on the index is 2.3%. What is the no-arbitrage price on a one-year index forward contract if the continuously compounded risk-free rate is 5%.

- A) 991.1.
 - B) 987.2.
 - C) 991.4.
-

Question #76 of 77

Question ID: 1473633

An index is currently 876, the risk-free rate (R_f) is 7%, and the dividend yield on the index portfolio is 1.8%. Assuming that these are continuously compounded yields, the price of an 18-month index future is *closest* to:

- A) 947.1.
 - B) 943.0.
 - C) 945.2.
-

Question #77 of 77

Question ID: 1473618

Which of the following *best* describes the price of a forward contract? The forward price is:

A) always equal to the market price at contract termination.

B) always expressed in dollars.

C) the price that makes the values of the long and short positions zero at contract initiation.